AJINKYA AWARI

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PROFILE

MSc Artificial Intelligence for Biomedicine and Healthcare student at UCL, building reproducible biomedical AI and evaluation systems with explicit evidence and limitations. Reports weak and null results openly.

EDUCATION

University College London (UCL), London, UK ***2026-2027***
MSc Artificial Intelligence for Biomedicine and Healthcare

Savitribai Phule Pune University, Pune, India | **CGPA: 8.25/10** ***2019-2023***
BE - Bachelor of Engineering, Computer Science. **Leadership:** Vice President, Technology Club

PROJECTS

AIXI-Complexity Benchmark

– To test how agent performance changes with task difficulty, built a local benchmark comparing five agent types (random, tabular Q-learning, an AIXI-inspired planner, a small LLM and an exact oracle) on 20 frozen finite-state tasks, with 81 tests and passing CI.

– Quantified results with hierarchical bootstrap intervals and Holm correction, and reported the null finding openly: no difficulty proxy remained significant (minimum adjusted $p = 0.144$).

GraphGPS-Upgrade

– To compare graph architectures fairly, ran GraphGPS, GCN and GIN under one frozen training budget on a Tesla T4, giving test ROC-AUC of 0.748, 0.693 and 0.698 respectively over three seeds; the result holds for this fixed recipe only.

ESM2 Protein Fitness Benchmark

– To compare protein language models fairly, built a leakage-aware ProteinGym benchmark with grouped splits (UniProt ID and sequence-similarity clusters), hashed split manifests and 112 tests.

– On 40 assays (CPU, small per-assay samples), ESM1v reached a macro Spearman of 0.475 against 0.266 for frozen ESM2-650M with ridge regression; confidence intervals await GPU access.

EXPERIENCE

Byju's Think & Learn Pvt. Ltd, India ***July 2023 - July 2024***
Cloud Engineer

– To help sales teams prioritise high-potential leads, implemented a predictive analytics pipeline in Python on cloud platforms, improving lead-to-customer conversion by 12% and driving an 8% increase in quarterly revenue.

– To keep the platform responsive during peak enrolments, redesigned cloud infrastructure (backend architecture, database queries and load balancing) for 300,000+ concurrent users, cutting latency by 15% and sustaining 99.7% uptime.

– To cut manual lead handling, integrated LeadSquared into the cloud-based lead-management workflow and refined it using cloud monitoring and performance metrics, saving 4+ hours a week and lifting lead conversion by 10% (contributing $300K+ in additional revenue).

The Leadership 30 ***July 2024 - Present***
Senior Data Analyst

– To improve emergency response in flood-prone districts of Maharashtra, built an AI-powered flood forecasting system with local agencies and assessed its hardware and software requirements, improving response times by 12% across six districts.

– To address learning gaps in rural public schools, led a three-month data-driven initiative with the Maharashtra government (SDG 4) and designed targeted interventions, raising student engagement by 15% and improving literacy outcomes for 2,000+ students.

– To inform solar micro-grid policy in rural Maharashtra, designed a cost-benefit model for renewable energy projects (SDG 7); the initiative led to a 10% rise in clean-energy adoption, benefiting 800+ households and cutting annual carbon emissions by 1,500 tons.

PUBLICATIONS

V. Paithankar, A. Awari (second author), A. Raskar, S. Patil, N. Jamdar. *Plant Disease Detection using Machine Learning.* IJARSCT, Vol. 3, Issue 2 (doi:10.48175/IJARSCT-9156) and Issue 4 (doi:10.48175/IJARSCT-9297), April 2023.

SKILLS

– **Languages:** Python, C, C++

– **ML / DL Frameworks:** PyTorch, TensorFlow, Keras, Scikit-learn

– **AI & Research Areas:** Graph Neural Networks, Protein language models, Federated learning, Model evaluation, Explainable AI, Reproducible research

– **Data Science Tools and Technologies:** Hadoop, Apache Spark, SQL, BigQuery, Docker, Matplotlib

– **Web & Tools:** HTML, CSS, Bootstrap, Flask, Streamlit, Git, AWS, GCP